Mock B · Paper 1: Wrong-Answer Notes

CFA Level II · 23 wrong answers, one per page

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Paper 1 (48%)

Score by topic

Topic	Score	%
Quant	0/4	0%
FI	1/4	25%
Corp	2/4	50%
FSA	2/4	50%
PM	4/8	50%
Alt	2/4	50%
Equity	5/8	62%
Ethics	5/8	62%

Questions at a glance

Q	Topic	You	Ans
Q3	Corp	B	C
Q4	Corp	C	A
Q6	FSA	A	B
Q8	FSA	A	B
Q9	PM	C	B
Q10	PM	C	A
Q12	PM	B	C
Q15	Equity	A	B
Q16	Equity	A	B
Q17	FI	B	A
Q19	FI	C	A
Q20	FI	C	A
Q22	Alt	C	A
Q24	Alt	C	B
Q25	Quant	B	C
Q26	Quant	B	A
Q27	Quant	B	C
Q28	Quant	A	C
Q31	Equity	C	B
Q35	PM	A	B
Q38	Ethics	C	A
Q40	Ethics	C	A
Q44	Ethics	B	A

Mock B · Paper 1 · Q3 You **B** · Answer **C**

Corp

VIGNETTE (RELEVANT PART, UNEDITED)

Singh would like to take advantage of the under-leveraged balance sheet of Axiom while ensuring that the company's current shareholders retain control. Singh considers what corporate actions could achieve these objectives.

Question 3. Which of the following corporate actions would best meet Singh's objectives?

OPTIONS

A	Reorganization		A reorganization is a court-supervised restructuring of an insolvent company. Axiom is under-leveraged and Singh wants to add debt, so this does not fit.
B	Leveraged buyout	✗ Your pick	An LBO is an acquirer-led strategy that uses a lot of debt to buy a target. Singh wants the current shareholders to keep control, so this does not fit.
C	Dividend recapitalization	✓ Correct	A dividend recapitalization restructures the mix from equity to debt through debt-financed dividends or repurchases. It replaces expensive equity with cheaper debt, does not dilute shareholders' stakes and can increase value to shareholders.

Trigger “under-leveraged balance sheet”
“current shareholders retain control”

Rule next time: Under-leveraged + shareholders must keep control + replace equity by debt = dividend recapitalization (debt-financed dividend or buyback). Reorganization = court-supervised insolvency; LBO = an acquirer buys the firm.

Key line: Dividend recap: debt up, equity down, stakes not diluted, WACC down.

Mock B · Paper 1 · Q4 You **C** · Answer **A**

Corp

VIGNETTE (RELEVANT PART, UNEDITED)

Division B generated an EBITDA of $10.8 million last year and is valued at 6.0x EV/EBITDA. Division B has two segments:

A wholesale segment which contributed 20% to the division's EBITDA. Comparable wholesale retailers are valued at 7.0x EV/EBITDA.

A consumer segment which contributed 80% to the division's EBITDA. Comparable consumer retailers are valued at 5.1x EV/EBITDA.

Singh evaluates if spinning off Division B's wholesale segment would have a material impact on shareholder value, which she defines as a change in shareholder value of greater than 5%.

Question 4. Based on Singh's analysis, spinning off the wholesale segment for Division B would lead to:

OPTIONS

A	a material decrease in shareholder wealth.	✓ Correct	Sum of the parts: 7.0 × 20% × 10.8 = 15.120 and 5.1 × 80% × 10.8 = 44.064, total 59.184. Division B is valued at 6 × 10.8 = 64.800, so the change is (59.184 − 64.800)/64.800 = −8.7%, a material decrease (the market values synergies between the segments).
B	no material impact on shareholder wealth.		Incorrect: this averages the two multiples ((7.0 + 5.1)/2 = 6.05), giving 65.340 and a change of only +0.8%.
C	a material increase in shareholder wealth.	✗ Your pick	Incorrect: this swaps the multiples (5.1 × 20% × 10.8 + 7.0 × 80% × 10.8 = 71.496), giving +10.3%.

Trigger “valued at 6.0x EV/EBITDA”
“contributed 20% to the division's EBITDA”
“change in shareholder value of greater than 5%”

Rule next time: Sum of the parts: each segment's EBITDA × ITS OWN comparable multiple, then compare with the combined EV. Material = change above ±5%.

Key line: SOTP = 7.0 × 20% × 10.8 + 5.1 × 80% × 10.8 = 59.184; combined = 6.0 × 10.8 = 64.8; change = −8.7%.

Mock B · Paper 1 · Q6 You **A** · Answer **B**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

On 1 January of the current year, Lonza granted 4 million stock options to employees. The options vest 3 years after the grant date and expire 6 years after the grant date. The fair value per option was EUR 1.8 on the grant date.

Question 6. For the current year, Lonza's compensation expense related to its stock options will be:

OPTIONS

A	EUR 1.2 million.	✗ Your pick	Incorrect: this divides the total fair value by the 6-year term to expiry instead of the vesting period.
B	EUR 2.4 million.	✓ Correct	Expense = (4 million × EUR 1.8) / 3-year vesting period = EUR 2.4 million.
C	EUR 7.2 million.		Incorrect: this recognizes the whole grant-date fair value in one year, without spreading it over the vesting period.

Trigger “vest 3 years after the grant date”
“expire 6 years after the grant date”

Rule next time: Option expense is spread over the VESTING period, not the term to expiry. One year's expense = grant-date fair value × options ÷ vesting years.

Key line: 4m × EUR1.8 ÷ 3 = EUR2.4m.

Mock B · Paper 1 · Q8 You **A** · Answer **B**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Exhibit 2

	Most Recent Year	Prior Year
Liquidity Coverage Ratio	95%	110%
Net Stable Funding Ratio	95%	90%

Question 8. Based on Lonza's liquidity ratios at the end of the most recent year, which of the following is correct?

OPTIONS

A	Lonza's Net Stable Funding Ratio has deteriorated from the prior year	✗ Your pick	Incorrect: a higher NSFR is an improvement and it rose from 90% to 95%.
B	Lonza's required stable funding is higher than its available stable funding	✓ Correct	NSFR = available stable funding / required stable funding. At 95% (below 100%), required stable funding is higher than available stable funding.
C	Lonza can withstand cash outflows 5% higher than its 30-day liquidity needs in a stress scenario		Incorrect: an LCR of 95% means Lonza can cover only 95% of its 30-day stress outflows, i.e. 5% less, not 5% more.

Trigger “Liquidity Coverage Ratio | 95% | 110%”
“Net Stable Funding Ratio | 95% | 90%”

Rule next time: NSFR = available ÷ required stable funding: below 100% means required > available; a higher NSFR than last year = improvement. LCR below 100% = cannot cover the 30-day stress outflows (it is short, not 5% to spare).

Key line: NSFR 95% < 100% → required stable funding is higher than available.

Mock B · Paper 1 · Q9 You **C** · Answer **B**

PM

VIGNETTE (RELEVANT PART, UNEDITED)

Grap backtests the strategy using a model with the following parameters:

• Rebalancing frequency: monthly (simulated using rolling windows);

• Formation of factor portfolios: for each factor, buy the top 20% of stocks and sell the bottom 20% of stocks ranked by the factor over the previous 12 months;

• Weighting of factor portfolios: risk parity.

Question 9. Which of the following will Grap's backtesting model require?

OPTIONS

A	Non-overlapping rolling windows		Incorrect: rolling windows overlap (each has a 12-month in-sample period, 11 months of which overlap with the previous window).
B	Two iterations of the rolling-window procedure	✓ Correct	One rolling-window iteration builds the factor portfolios and a second one over the same time span estimates the covariance matrix for risk parity weights, which avoids look-ahead bias.
C	A 13-month in-sample period for each rolling window	✗ Your pick	Incorrect: the in-sample period is the previous 12 months (not 13) with a 1-month out-of-sample period.

Trigger "Rebalancing frequency: monthly (simulated using rolling windows)"
"Weighting of factor portfolios: risk parity"

Rule next time: Two layers (factor portfolios, then risk-parity weights from a covariance matrix) = TWO rolling-window iterations; windows overlap; the in-sample period is the previous 12 months (not 13).

Key line: Layer 1: factor ranks over 12 months. Layer 2: 10-year covariance matrix → equal-risk-contribution weights.

Mock B · Paper 1 · Q10 You **C** · Answer **A**

PM

VIGNETTE (RELEVANT PART, UNEDITED)

The TOP100 Index is updated regularly and, as a result, its composition has significantly changed over the 10-year backtesting period. Grap wants to ensure that survivorship bias is mitigated and compares three alternative methods to form factor portfolios in each rolling window:

- Method A only uses the companies that were in the TOP100 Index at any given point in time t.
- Method B only uses the companies that were in the TOP100 Index at the end of the 10-year period.
- Method C only uses the companies that have been in the TOP100 Index for the entire 10-year period.

Question 10. To mitigate survivorship bias, which method should Grap use when forming factor portfolios?

OPTIONS

A	Method A	✓ Correct	Point-in-time data: in each rolling window use only the companies that were in the index at that time (casualties as well as survivors).
B	Method B		Incorrect: only companies in the index at the end of the period are survivors, which creates survivorship bias.
C	Method C	✗ Your pick	Incorrect: only companies in the index for the entire 10 years are survivors, which creates survivorship bias.

Trigger “composition has significantly changed”
“Method A only uses the companies that were in the TOP100 Index at any given point in time t.”

Rule next time: Survivorship bias is avoided with point-in-time data (members as they were at each date, including later casualties). Methods that start from today's members, or from names present throughout, keep only survivors.

Key line: Point-in-time universe = no survivorship bias.

Mock B · Paper 1 · Q12 You **B** · Answer **C**

PM

VIGNETTE (RELEVANT PART, UNEDITED)

Grap performs a complementary risk analysis using a risk measure that can give him a better understanding of the downside risk of the absolute return trading strategy to be employed by the equity market-neutral fund. The strategy is benchmarked against cash.

Question 12. For his complementary risk analysis, which of the following risk measures should Grap use?

OPTIONS

A	Relative VaR		Incorrect: with a cash benchmark (zero volatility) relative VaR equals VaR, so it adds no information.
B	Surplus at risk	✗ Your pick	Incorrect: surplus at risk is for pension funds with liabilities, not for a hedge fund.
C	Maximum drawdown	✓ Correct	For an absolute return, equity market-neutral strategy, maximum drawdown (worst peak-to-trough decline) complements standard deviation and beta as a downside risk measure.

Trigger “absolute return trading strategy”
“benchmarked against cash”
“better understanding of the downside risk”

Rule next time: Absolute return vs a cash benchmark: relative VaR equals plain VaR (adds nothing); surplus at risk is for pension liabilities; downside of a market-neutral strategy = maximum drawdown (worst peak-to-trough).

Key line: Cash benchmark has zero volatility → relative VaR = VaR.

Mock B · Paper 1 · Q15 You **A** · Answer **B**

Equity

VIGNETTE (RELEVANT PART, UNEDITED)

Kassam evaluates the stock of Altair, which is a recently listed, high growth, biotechnology company. She is considering a stable market-based multiple to value the stock and does not want to make any adjustments to the reported financial statements. After reviewing recent analyst reports, Kassam notices that Altair has internally generated brands that are not shown as intangible assets on Altair's balance sheet. Further, she notes that certain discretionary accounting decisions about expenses have led to an inaccurate representation of Altair's economic performance.

Question 15. Which of the following valuation ratios is most suitable for Kassam to value Altair?

OPTIONS

A	Price-to-book ratio	✗ Your pick	Incorrect: P/B misses internally generated intangibles such as brands, which are not on the balance sheet.
B	Price-to-sales ratio	✓ Correct	Sales are less subject to distortion or manipulation than EPS or book value and are more stable than EPS, so P/S is suitable without adjusting reported financials.
C	Price-to-earnings ratio		Incorrect: earnings are prone to manipulation through discretionary expense decisions and distort EPS.

Trigger “stable market-based multiple”
“does not want to make any adjustments to the reported financial statements”
“internally generated brands”

Rule next time: No adjustments wanted + unreliable earnings and book value (internal brands, discretionary expenses) → P/S: sales are hard to manipulate and more stable than EPS. P/B misses internal intangibles; P/E is distorted by expense choices.

Key line: P/S: stable, less manipulable, no adjustments needed.

Mock B · Paper 1 · Q16 You **A** · Answer **B**

Equity

VIGNETTE (RELEVANT PART, UNEDITED)

Kassam values Becq, a small private company, using the excess earnings method. Relevant financial information is shown in Exhibit 3.

Exhibit 3

Fixed assets	$725,000
Working capital	$315,000
Last reported earnings	$191,290
Residual income growth rate	2.50%
Required return on fixed assets	8.50%
Required return on working capital	3.25%
Discount rate	11.25%

Question 16. Based on Exhibit 3, the value of Becq using the excess earnings method is closest to:

OPTIONS

A	$1,400,000.	✗ Your pick	Incorrect: this is only the value of the intangibles (1,399,008); tangible assets must be added.
B	$2,440,000.	✓ Correct	Residual earnings = 191,290 − 315,000 × 3.25% − 725,000 × 8.50% = 119,427.50. Intangibles = 119,427.50 × 1.025/(11.25% − 2.50%) = 1,399,008. Total = 1,399,008 + 315,000 + 725,000 = about 2,440,000.
C	$3,080,000.		Incorrect: the required return on fixed assets (8.50%) is used as the capitalization rate instead of the discount rate.

Trigger “excess earnings method”
“Required return on working capital | 3.25%”
“Discount rate | 11.25%”

Rule next time: Excess earnings: residual = earnings − required returns on working capital and fixed assets; capitalise it with the DISCOUNT rate − g (not the asset return) to value the intangibles, then ADD the tangible assets for the whole business.

Key line: 119,427.5 × 1.025 ÷ (11.25% − 2.5%) = 1,399,008; + 315,000 + 725,000 = 2.44m.

Mock B · Paper 1 · Q17 You **B** · Answer **A**

FI

VIGNETTE (RELEVANT PART, UNEDITED)

She first considers the following 2-year, default-free bonds with embedded options:

Bond A is a 5% annual pay coupon bond and is callable at par ($100) at the end of year 1.

To analyze these bonds, Plessinger creates a binomial interest rate tree using the rates in Exhibit 1, assuming an interest rate volatility of 20%, with branching from each node to a high state and a low state occurring with equal probability. Plessinger also considers the effective convexity of Bond B.

Exhibit 1 Binomial Interest Rate Tree (1-year rates)

Year 0	Year 1	Year 2	
Upper node		6.09%	7.32%
Middle node	2.00%		4.91%
Lower node		4.08%	3.29%

Note: This exhibit was a binomial diagram in the original paper. Its figure was not in the pasted text: the Year 0 and Year 1 rates come from the answer explanations, the Year 2 nodes are reconstructed from them (4.08% × e^0.4 = 6.09%, 3.29% × e^0.4 = 4.91% and 3.29% × e^0.8 = 7.32%).

Question 17. Based on the binomial interest rate tree, the value of Bond A is closest to:

OPTIONS

A	$102.44.	✓ Correct	Work backwards: at 6.09% the value is 105/1.0609 = 98.97; at 4.08% it is 105/1.0408 = 100.88, which exceeds par so the bond is called at 100. Year 0: (100 + 5)/1.02 = 102.9412 and (98.97 + 5)/1.02 = 101.9339; the average is 102.44.
B	$102.87.	✗ Your pick	Incorrect: this ignores the call (uses 100.88 instead of 100): 102.87.
C	$103.37.		Incorrect: this treats the bond as putable instead of callable (103.37).

Trigger “Bond A is a 5% annual pay coupon bond and is callable at par”
“interest rate volatility of 20%”

Rule next time: Callable bond on the tree: work backwards from the end; at any node where the value exceeds the call price (100) the issuer calls, so cap the value at 100; then average the two nodes and discount with the node rate.

Key line: Year 1: 105/1.0609 = 98.97 and min(105/1.0408, 100) = 100; (100+5)/1.02 and (98.97+5)/1.02 → 102.44.

Mock B · Paper 1 · Q19 You **C** · Answer **A**

FI

VIGNETTE (RELEVANT PART, UNEDITED)

Plessinger next evaluates a 3-year maturity, callable investment-grade corporate bond. She computes an OAS of 51 bps using the same binomial interest rate tree in Exhibit 1. Plessinger considers the impact on the bond's OAS under a scenario where the bond's price is unchanged and interest rate volatility increases from 20% to 25%.

Question 19. Under Plessinger's scenario, the OAS of the corporate bond will:

OPTIONS

A	decrease.	✓ Correct	Higher volatility raises the value of the embedded call. With the price (and Z-spread) fixed, and Z-spread = OAS + option cost, the OAS must decline.
B	not change.		Incorrect: higher volatility decreases the OAS of a callable bond.
C	increase.	✗ Your pick	Incorrect: higher volatility decreases the OAS of a callable bond.

Trigger “callable investment-grade corporate bond”
“interest rate volatility increases from 20% to 25%”
“the bond's price is unchanged”

Rule next time: Callable bond, price fixed, volatility up: the call option is worth more; Z-spread = OAS + option cost, so the OAS FALLS (for a putable bond it would rise).

Key line: Z-spread = OAS + option cost.

Mock B · Paper 1 · Q20 You **C** · Answer **A**

FI

VIGNETTE (RELEVANT PART, UNEDITED)

Exhibit 1 Binomial Interest Rate Tree (1-year rates)

Year 0	Year 1	Year 2	
Upper node		6.09%	7.32%
Middle node	2.00%		4.91%
Lower node		4.08%	3.29%

Note: This exhibit was a binomial diagram in the original paper. Its figure was not in the pasted text: the Year 0 and Year 1 rates come from the answer explanations, the Year 2 nodes are reconstructed from them (4.08% × e^0.4 = 6.09%, 3.29% × e^0.4 = 4.91% and 3.29% × e^0.8 = 7.32%).

Finally, still using the binomial interest rate tree in Exhibit 1, Plessinger analyzes a 3-year, default-free capped floating rate bond with a par value of $100. The bond pays an annual coupon equal to the 1-year reference rate, is set in arrears, and is capped at 5.0%.

Question 20. The value of the capped floating rate bond analyzed by Plessinger is:

OPTIONS

A	less than its par value.	✓ Correct	The cap of 5% binds when rates are above 5% (the 6.09% Year 1 rate and the 7.32% Year 2 rate), so there are paths where the floater's coupon is limited and its value is below par.
B	equal to its par value.		Incorrect: without the cap a floater sells at par on each reset date, but the cap binds on some paths in the tree even though the spot rates are below 5%.
C	greater than its par value.	✗ Your pick	Incorrect: a cap can only reduce the value of a floater.

Trigger “capped floating rate bond”
“capped at 5.0%”
“set in arrears”

Rule next time: A cap limits the coupon on paths where the tree rate exceeds the cap (6.09% in Year 1, 7.32% in Year 2), so the capped floater is worth LESS than par; without a cap a floater trades at par on reset dates.

Key line: Cap binds on some nodes → value < par.

Mock B · Paper 1 · Q22 You **C** · Answer **A**

Alt

VIGNETTE (RELEVANT PART, UNEDITED)

Chang is also evaluating hedge funds as potential additions to the portfolio. She identifies the following strategies and considers their returns, leverage, and liquidity:

Strategy 1: Multi-strategy funds

Strategy 2: Fund of funds

Question 22. Compared to Strategy 2, Strategy 1 most likely has:

OPTIONS

A	higher returns.	✓ Correct	Multi-strategy funds have generally outperformed funds of funds, but with more variance and occasional large losses linked to higher leverage; flexibility and fast tactical allocation help.
B	lower leverage.		Incorrect: multi-strategy funds use significantly more leverage than most funds of funds.
C	fewer gates on redemptions.	✗ Your pick	Incorrect: multi-strategy funds often impose investor-level or fund-level gates, unlike funds of funds.

Trigger “Strategy 1: Multi-strategy funds”
“Strategy 2: Fund of funds”
“returns, leverage, and liquidity”

Rule next time: Multi-strategy vs fund of funds: multi-strategy = higher returns, MORE leverage, more variance and more gates; funds of funds = more diversified mix, lower leverage, somewhat lower variance.

Key line: Multi-strategy: return up, leverage up, gates yes.

Mock B · Paper 1 · Q24 You **C** · Answer **B**

Alt

VIGNETTE (RELEVANT PART, UNEDITED)

The pension plan has an objective to limit drawdowns. Chang considers what individual hedge fund strategy would best meet that objective when combined with the pension plan's stock and bond portfolio.

Question 24. Which of the following hedge fund strategies is best suited to meet the plan's drawdown objective?

OPTIONS

A	Event-driven		Incorrect: event-driven/distressed strategies show high maximum drawdowns when combined with a stock and bond portfolio and little risk reduction.
B	Global macro	✓ Correct	The opportunistic strategies (global macro, systematic futures, merger arbitrage, equity market-neutral) give the smallest maximum drawdowns when combined with a traditional portfolio and have low serial autocorrelation (good liquidity).
C	Relative value	✗ Your pick	Incorrect: relative value/convertible arbitrage strategies show high maximum drawdowns and high standard deviations when combined.

Trigger “limit drawdowns”
“combined with the pension plan's stock and bond portfolio”

Rule next time: Smallest maximum drawdown when added to a stock and bond portfolio: opportunistic strategies (global macro, systematic futures, merger arbitrage, equity market-neutral). Event-driven, relative value and long/short equity show high drawdowns.

Key line: Low drawdown contribution → global macro.

Mock B · Paper 1 · Q25 You **B** · Answer **C**

Quant

VIGNETTE (RELEVANT PART, UNEDITED)

Moreno is tasked with a new project that looks to classify bond issuers based on their probability of default within the next 12 months. He gathers a large sample of financial and accounting metrics for 1,000 issuers and observes that the issuers are not linearly separable. He compares non-linear support vector machine (SVM), unconstrained classification and regression tree (CART), and SVM with soft margin classification. He wants to use a machine-learning algorithm that addresses non-linear separability and is not prone to overfitting.

Question 25. Which of the following machine-learning algorithms is most suitable for Moreno's bond issuer classification model?

OPTIONS

A	Non-linear SVM		Incorrect: non-linear SVM with more advanced boundaries reduces training misclassifications but is complex and prone to overfitting.
B	Unconstrained CART	✗ Your pick	Incorrect: unconstrained CART can perfectly learn the training data and overfit unless regularization parameters are added.
C	SVM with soft margin classification	✓ Correct	SVM with soft margin classification adds a penalty for misclassified training observations and trades off a wider margin against total error, handling non-linear separability without overfitting.

Trigger “not linearly separable”
“addresses non-linear separability and is not prone to overfitting”

Rule next time: Non-linear separability + no overfitting → SVM with soft margin classification (penalty for misclassified points, wider margin). Non-linear SVM and unconstrained CART overfit.

Key line: Soft margin: trade-off wider margin vs total error penalty.

Mock B · Paper 1 · Q26 You **B** · Answer **A**

Quant

VIGNETTE (RELEVANT PART, UNEDITED)

Moreno also creates Model 1, a logistic regression model that uses two factors, profit margin (PROFIT) and leverage ratio (LEVER), to predict the probability of default. The binary dependent variable is 1 for companies that defaulted within the subsequent 12 months and 0 otherwise. Using the regression results in Exhibit 1, Moreno calculates the predicted probability of default for the average distressed company, which has PROFIT = –2.0 and LEVER = 3.0.

Exhibit 1

	Coefficient
Intercept	–1.7878
PROFIT	–0.0185
LEVER	0.3876

Question 26. For the average distressed company, the probability of default predicted by Model 1 is closest to:

OPTIONS

A	36%.	✓ Correct	P = 1/[1 + exp(−(−1.7878 − 0.0185 × (−2.0) + 0.3876 × 3.0))] = 1/(1 + 1.8004) = 0.357, about 36%.
B	56%.	✗ Your pick	Incorrect: 0.56 is exp(b0 + b1PROFIT + b2LEVER) = exp(−0.5880), the odds, not the probability.
C	77%.		Incorrect: 77% omits the intercept.

Trigger “PROFIT = –2.0 and LEVER = 3.0”
“Intercept | –1.7878”

Rule next time: Logit: P = 1 ÷ (1 + exp(−(b0 + b1x1 + b2x2))). exp(linear part) alone is the ODDS, not the probability, and dropping the intercept gives 77%.

Key line: x = −1.7878 + 0.0370 + 1.1628 = −0.588; P = 1 ÷ (1 + e^0.588) = 0.357.

Mock B · Paper 1 · Q27 You **B** · Answer **C**

Quant

VIGNETTE (RELEVANT PART, UNEDITED)

To analyze the drivers of ROE for software companies, Moreno collects fundamental data and compares two linear regression models. Model 2 uses profit margin (PROFIT) and interest coverage (COVER). Model 3 extends Model 2 by adding a third factor, the quick ratio (QUICK).

Exhibit 2

Coefficient	t-Statistic	p-Value	
Intercept	–1.4411	–3.4888	0.0000
PROFIT	0.4840	49.5319	0.0000
COVER	0.7487	27.9215	0.0000
QUICK	0.0919	1.9204	0.0558

Question 27. Compared to Model 2, Model 3 most likely has a:

OPTIONS

A	lower R-squared and a higher adjusted R-squared.		Incorrect: R^2 never decreases when a variable is added.
B	higher R-squared and a lower adjusted R-squared.	✗ Your pick	Incorrect: adjusted R^2 rises when the added coefficient's $\|t\| > 1.0$, and QUICK's t-statistic is 1.9204.
C	higher R-squared and a higher adjusted R-squared.	✓ Correct	R^2 increases or stays the same when a variable is added, and adjusted R^2 increases if the new coefficient's |t-statistic| is above 1.0 (QUICK: 1.9204).

Trigger “adding a third factor, the quick ratio (QUICK)”
“QUICK | 0.0919 | 1.9204 | 0.0558”

Rule next time: Adding a variable never lowers R^2; adjusted R^2 rises if the new coefficient's $|t| > 1.0$ (QUICK: 1.92), even though p = 0.0558 is not significant at 5%.

Key line: $|t| > 1 \rightarrow$ adjusted R^2 up; $|t| < 1 \rightarrow$ down.

Mock B · Paper 1 · Q28 You **A** · Answer **C**

Quant

VIGNETTE (RELEVANT PART, UNEDITED)

Finally, Moreno wants to test if last-12-month sales (SALES) are a good predictor of ROE. He runs Model 4, which has the following specification:

A plot of the regression residuals against SALES is presented in Exhibit 3.

Note: Exhibit 3 was an image in the original paper (file KL3.png) and was not in the pasted text. From the answer explanation: in the plot of the Model 4 residuals against SALES, the variance of the residuals increases as SALES increase.

Question 28. The most appropriate conclusion to draw from the residual plot for Model 4 is that:

OPTIONS

A	the model suffers from unconditional heteroskedasticity.	✗ Your pick	Incorrect: the residual variance increases with SALES, so it is correlated with an independent variable: conditional, not unconditional, heteroskedasticity.
B	there is a non-linear relationship between ROE and SALES.		Incorrect: a residual plot does not show the form of the relationship between target and feature.
C	the F-test for the overall regression significance is unreliable.	✓ Correct	With conditional heteroskedasticity the MSE is a biased estimator, so the F-test for overall significance is unreliable (and Type I errors rise).

Trigger “A plot of the regression residuals against SALES”
“the variance of the residuals increases as SALES increase”
Rule next time: Residual variance that grows with the regressor = CONDITIONAL heteroskedasticity: biased standard errors, unreliable F-test, more Type I errors. Unconditional heteroskedasticity is not a problem. A residual plot does not show non-linearity of the relationship.

Key line: Fan-shaped residuals → conditional heteroskedasticity.

Mock B · Paper 1 · Q31 You **C** · Answer **B**

Equity

VIGNETTE (RELEVANT PART, UNEDITED)

McCormick reviews his FCFE valuation model for Firm C. Firm C's management has just announced that they will issue new common stock during the next year. McCormick evaluates the potential effect of this change on next year's FCFE forecast.

Question 31. Given Firm C's announced change, next year's FCFE forecast will most likely:

OPTIONS

A	decrease.		Incorrect: transactions between the company and its shareholders (dividends, repurchases, issuances) do not affect free cash flow.
B	remain unchanged.	✓ Correct	Share issuances are a use of FCFE, not a determinant of it, so the forecast is unchanged.
C	increase.	✗ Your pick	Incorrect: share issuance does not affect free cash flow.

Trigger “will issue new common stock during the next year”
“next year's FCFE forecast”

Rule next time: Share issuance, repurchases and dividends are USES of FCFE, not drivers of it: next year's FCFE is unchanged. FCFE changes only with CFO, capex and net borrowing.

Key line: FCFE = CFO − FCInv + net borrowing.

Mock B · Paper 1 · Q35 You **A** · Answer **B**

PM

VIGNETTE (RELEVANT PART, UNEDITED)

Chu reviews a diversified US bond portfolio, Portfolio A. His objective is to determine which of the following three sectors' spreads can be expected to narrow the most if the economy recovers from a business downturn.

Sector A: Investment grade consumer cyclical

Sector B: Speculative grade consumer cyclical

Sector C: Speculative grade consumer noncyclical

Question 35. For Portfolio A, which of the following sectors most likely satisfies Chu's objective concerning credit spreads?

OPTIONS

A	Sector A	✗ Your pick	Incorrect: investment grade cyclical spreads narrow less than speculative grade cyclical spreads.
B	Sector B	✓ Correct	Sector B is both speculative grade and cyclical, so it is the most sensitive to an improving economy and its spreads narrow the most.
C	Sector C		Incorrect: noncyclical issuers are less sensitive to the business cycle than cyclical issuers.

Trigger “spreads can be expected to narrow the most”
“economy recovers from a business downturn”
Rule next time: Spreads move most for low-rated AND cyclical issuers: they widen most in a downturn and narrow most in a recovery → speculative grade + cyclical = Sector B.

Key line: Low rating + cyclical = biggest spread swings.

Mock B · Paper 1 · Q38 You **C** · Answer **A**

Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

Silva conducts due diligence on a mid-cap packaging company whose stock is actively traded. He determines the company's fundamentals are strong and buys its stock for one of the equity funds he manages. Silva provides commentary on the company's prospects in his monthly client newsletter. Silva has owned the company's stock in his personal portfolio for many years and has disclosed this in the fund's communications to clients. A few weeks after releasing the newsletter, Silva sells some of his personal position in the stock as part of rebalancing his personal portfolio. He earns a substantial profit from this sale.

Question 38. Did Silva violate the Standards when selling his personal stake in the packaging company?

OPTIONS

A	No	✓ Correct	Standard VI(B) Priority of Transactions: client transactions have priority. The fund bought first and selling an actively traded stock from his personal portfolio did not disadvantage clients.
B	Yes, because he sold the shares at a profit		Incorrect: a profit is not a violation as long as clients are not disadvantaged.
C	Yes, by acting against his own recommendation	✗ Your pick	Incorrect: the Standard does not prohibit personal transactions that differ from current recommendations as long as clients are not disadvantaged.

Trigger “buys its stock for one of the equity funds he manages”
“Silva sells some of his personal position”
“as part of rebalancing his personal portfolio”

Rule next time: VI(B) Priority of Transactions: client trades come first; personal trades are allowed if clients are not disadvantaged (the fund bought first, the stock is actively traded). A profit, or trading against his own recommendation, is not a violation by itself.

Key line: No client disadvantage → no violation.

Mock B · Paper 1 · Q40 You **C** · Answer **A**

Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

Tritium's recommended policy for record retention is to maintain records of investment actions and client communications in electronic format. Silva is accustomed to retaining all of his and his clients' relevant files in hard copy form. He does not store any of the records in electronic format. He saves his records for 5 years, after which they are destroyed. Local regulations require record retention for 5 years.

Question 40. Does Silva violate the Standards with respect to record retention?

OPTIONS

A	No	✓ Correct	Standard V(C): records may be kept in hard copy or electronic form, and following the regulatory requirement (5 years) satisfies the Standard; 7 years applies only absent regulation or firm policy.
B	Yes, because of the format in which the records are kept		Incorrect: the Standard does not require a specific format.
C	Yes, because records must be retained for a minimum period of 7 years	✗ Your pick	Incorrect: the 7-year recommendation applies only if there is no regulatory requirement; the local rule is 5 years.

Trigger “hard copy form”
“He saves his records for 5 years”
“Local regulations require record retention for 5 years”

Rule next time: V(C): records may be kept in hard copy or electronic form, for the period required by regulation or firm policy; the 7-year recommendation applies only when neither exists.

Key line: Format free; period = regulation (5 years) ≥ OK.

Mock B · Paper 1 · Q44 You **B** · Answer **A**

Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

Murphy's supervisor manages a large research department and, as a result, is not able to personally evaluate the conduct of all the employees in the department. The supervisor hires a deputy manager to oversee several employees and ensures the deputy manager is properly trained. The supervisor establishes the following policies to assist with compliance:

Policy 1: A monetary incentive will be offered as a reward for ethical behavior.

Policy 2: The deputy manager will be responsible for evaluating the ethical conduct of department members.

Question 44. Does Murphy's supervisor violate the Standards in his oversight of the department?

OPTIONS

A	No	✓ Correct	Standard IV(C): incentives (monetary or not) that reward ethical behavior are allowed and delegating supervisory duties in a large department is reasonable.
B	Yes, by establishing Policy 1	✗ Your pick	Incorrect: monetary incentives for ethical behavior do not violate the Standard.
C	Yes, by establishing Policy 2		Incorrect: delegating supervision to a properly trained deputy is not a violation.

Trigger “A monetary incentive will be offered as a reward for ethical behavior.”
“The deputy manager will be responsible for evaluating the ethical conduct of department members.”

Rule next time: IV(C): a supervisor of a large department may delegate to trained deputies and may reward ethical behavior, even with money; responsibility remains with the supervisor.

Key line: Incentive for ethics + trained delegate = compliant.

Mock B · Paper 2: Wrong-Answer Notes

CFA Level II · 19 wrong answers, one per page

25 / 44

Paper 2 (57%)

Score by topic

Topic	Score	%
Econ	1/4	25%
Ethics	1/4	25%
Quant	2/4	50%
PM	2/4	50%
Equity	2/4	50%
FSA	5/8	62%
FI	5/8	62%
Alt	3/4	75%
Deriv	4/4	100%

Questions at a glance

Q	Topic	You	Ans
Q1	FSA	A	B
Q3	FSA	A	B
Q5	Quant	C	A
Q6	Quant	B	A
Q10	Econ	B	C
Q11	Econ	C	A
Q12	Econ	B	C
Q15	PM	B	C
Q16	PM	B	C
Q18	FI	C	A
Q20	FI	A	B
Q21	FSA	C	B
Q25	Equity	C	A
Q26	Equity	C	B
Q32	FI	A	B
Q38	Alt	A	B
Q41	Ethics	A	C
Q42	Ethics	C	B
Q43	Ethics	C	A

Mock B · Paper 2 · Q1 You **A** · Answer **B**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Tay collects components of Buttery's net sales growth from various markets of the most recent year in Exhibit 1.

Exhibit 1

Market	Volume Growth	Price Growth	Foreign Currency Exchange
Thailand	5%	0%	4%
Malaysia	4%	5%	–4%
Hong Kong SAR	–4%	8%	0%

Question 1. The market with the highest sustainable sales growth rate is:

OPTIONS

A	Thailand.	✗ Your pick	Incorrect: Thailand's organic growth is only 5% + 0% = 5%; 4% came from exchange rates, which is less sustainable.
B	Malaysia.	✓ Correct	Sustainable growth comes from volume and price, not exchange rates. Malaysia: 4% + 5% = 9%, above Thailand (5%) and Hong Kong SAR (−4% + 8% = 4%).
C	Hong Kong SAR.		Incorrect: Hong Kong SAR's volume and price growth is −4% + 8% = 4%, lower than Malaysia's 9%.

Trigger “Volume Growth | Price Growth | Foreign Currency Exchange”
“highest sustainable sales growth rate”

Rule next time: Sustainable sales growth = volume + price; the foreign-exchange part is not sustainable.

Key line: Malaysia 4 + 5 = 9%; Thailand 5 + 0 = 5%; Hong Kong −4 + 8 = 4%.

Mock B · Paper 2 · Q3 You **A** · Answer **B**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Creamy Meadows is a subsidiary of Buttery. Creamy began operations in Indonesia on 1 January of the most recent year and uses the Indonesian Rupiah (IDR) as its functional currency. Buttery uses the temporal method to translate Creamy's financial results. For the most recent year, Creamy had a net monetary liability and the IDR strengthened against the USD. Tay believes that the translation method may distort the financial and operating measures of the international subsidiary. To verify this, Tay compares Creamy's translated ratios to the ratios calculated under the local currency by using the exchange rates of the most recent year in Exhibit 2

Exhibit 2

Date	IDR per USD (Amount of IDR per 1 USD)
1 January of the most recent year	15,000
Average for the most recent year	14,700
Weighted average rate when inventory was purchased	14,700
31 December of the most recent year	14,000

Question 3. Which of the following ratios for Creamy most likely remains unchanged after translation for the most recent year?

OPTIONS

A	Current ratio	✗ Your pick	Incorrect: current assets include inventory translated at the historical rate (14,700) and monetary items at the year-end rate (14,000), while liabilities are at 14,000, so the ratio changes.
B	COGS to sales ratio	✓ Correct	Under the temporal method sales are translated at the average rate and COGS at the rate when the inventory was purchased; both are 14,700, so the translated ratio equals the local-currency ratio.
C	Sales to receivables ratio		Incorrect: sales use the average rate (14,700) and receivables the year-end rate (14,000), so the ratio changes.

Trigger “Buttery uses the temporal method”
“Weighted average rate when inventory was purchased | 14,700”
“Average for the most recent year | 14,700”

Rule next time: Temporal method: monetary items at the current rate, inventory/COGS at historical rates, sales at the average rate. A ratio is unchanged only if numerator and denominator are translated at the same rate (COGS ÷ sales: 14,700 for both).

Key line: COGS (14,700) ÷ sales (14,700): same rate.

Mock B · Paper 2 · Q5 You **C** · Answer **A**

Quant

VIGNETTE (RELEVANT PART, UNEDITED)

Danny Aoun is an equity portfolio manager at Betterflow Investments. He is developing sentiment indicators for forecasting stock price movements. Aoun gathers textual data from the internet to build machine-learning models. He extracts raw content from a variety of online sources and then trains the models for sentiment prediction. Each text item is labelled as either positive sentiment or negative sentiment. Aoun is more concerned about the models producing false negatives than false positives. He computes selected performance metrics for three models using confusion matrices and presents the metrics in Exhibit 1.

Exhibit 1

Model	Precision	Recall	F1 Score	Accuracy
X	0.87	0.94	0.90	0.88
Y	0.93	0.90	0.91	0.92
Z	0.96	0.82	0.88	0.86

Question 5. Which machine-learning model should Aoun choose given his concern about false negatives?

OPTIONS

A	Model X	✓ Correct	Recall (correctly predicted positives / all actual positives) is the metric to use when false negatives are costly. Model X has the highest recall (0.94).
B	Model Y		Incorrect: Model Y has the highest accuracy and F1 score, which weight false positives and false negatives equally.
C	Model Z	✗ Your pick	Incorrect: Model Z has the highest precision, which matters when false positives are costly.

Trigger “more concerned about the models producing false negatives”
“Precision | Recall | F1 Score | Accuracy”

Rule next time: False negatives costly → highest RECALL (TP ÷ (TP + FN)); false positives costly → precision; equal weight → accuracy or F1.

Key line: Recall: X 0.94 > Y 0.90 > Z 0.82.

Mock B · Paper 2 · Q6 You **B** · Answer **A**

Quant

VIGNETTE (RELEVANT PART, UNEDITED)

For Bytewave, Aoun estimates a first-order autoregressive model (AR(1)) on the stock price. From the results, he concludes that the time-series model exhibits autoregressive conditional heteroskedasticity and has ARCH(1) errors.

Question 6. Based on Aoun's conclusion regarding the time-series model for Bytewave, the variance of the error term is:

OPTIONS

A	not constant but predictable.	✓ **Correct**	With ARCH(1) errors the variance in period t + 1 depends on the squared error of period t, so it is not constant but predictable.
B	both constant and predictable.	✗ **Your pick**	Incorrect: ARCH errors have a non-constant variance.
C	neither constant nor predictable.		Incorrect: ARCH(1) makes the variance depend linearly on the previous squared error, so it is predictable.

Trigger “autoregressive conditional heteroskedasticity”
“ARCH(1) errors”

Rule next time: ARCH(1): the error variance depends on the previous period's squared error, so it is NOT constant but predictable.

Key line: $\text{Var}(\varepsilon_{t+1}) = a_0 + a_1 \cdot \varepsilon_t^2$.

Mock B · Paper 2 · Q10 You **B** · Answer **C**

Econ

VIGNETTE (RELEVANT PART, UNEDITED)

Team Green is asked to discuss how investment in physical capital, human capital, and technological development affects economic growth. As a summary of the discussion, Team Green presents a number of points including the following three:

Point 1: Since technology results in an upward shift in the production function, the relationship between economic growth and R&D spending is positive in the short run.

Point 2: Total factor productivity reflects the state of technology, and can be measured by the contribution of capital deepening plus the growth rates of labor productivity.

Point 3: Although long-term sustainable growth cannot rely on pure capital deepening, the impact of investment spending on available capital depends on the existing physical capital stock.

Question 10. Which of the points raised by Team Green is most likely correct?

OPTIONS

A	Point 1		Incorrect: the relationship between R&D spending and growth is not clear-cut in the short run (creative destruction can slow growth cyclically).
B	Point 2	✗ Your pick	Incorrect: labor productivity growth equals the contribution of capital deepening plus TFP growth, so TFP is not capital deepening plus labor productivity growth.
C	Point 3	✓ Correct	The impact of investment on available capital depends on the existing capital stock (large in developing countries), even though sustainable long-term growth cannot rely on capital deepening alone.

Trigger “Total factor productivity reflects the state of technology”
“the impact of investment spending on available capital depends on the existing physical capital stock”

Rule next time: Labor productivity growth = capital deepening + TFP growth, so TFP is NOT 'capital deepening plus labor productivity'. The R&D–growth link is not clear-cut in the short run. Investment matters most where the capital stock is small.

Key line: Labor productivity growth = capital deepening + TFP growth.

Mock B · Paper 2 · Q11 You **C** · Answer **A**

Econ

VIGNETTE (RELEVANT PART, UNEDITED)

Team Purple is asked to investigate whether the per capita income of a developing country should converge toward that of developed countries over time. Team Purple reviews various determinants of per capital incomes: saving rate, population growth rate, the shape of production function, and the externalities associated with human capital. After a debate among team members, Team Purple concludes that the externalities associated with human capital is the most important determinant in predicting the occurrence of convergence.

Question 11. Which of the following convergence concepts is most compatible with Team Purple's conclusion?

OPTIONS

A	Non-convergence	✓ Correct	Externalities from human capital (endogenous growth) remove diminishing returns, so rich countries can keep their lead; the model makes no prediction that convergence occurs.
B	Absolute convergence		Incorrect: absolute convergence assumes diminishing marginal returns to capital and the same technology.
C	Conditional convergence	✗ Your pick	Incorrect: conditional convergence also assumes diminishing returns to capital (the neoclassical model).

Trigger “externalities associated with human capital”
“most important determinant in predicting the occurrence of convergence”

Rule next time: Human capital and knowledge externalities = endogenous growth: no diminishing returns, so rich countries keep their lead → NON-convergence. Diminishing returns (neoclassical) → conditional or absolute convergence.

Key line: Externalities → no convergence.

Mock B · Paper 2 · Q12 You **B** · Answer **C**

Econ

VIGNETTE (RELEVANT PART, UNEDITED)

Smith presents the expected impact of removing trade barriers to economic growth. Referring to the neoclassical growth model, Smith concludes that opening up the economy should increase the rate at which countries' capital-to-labor ratios converge. Team Orange is asked to describe the dynamic adjustment process consistent with Smith's conclusion. Team Orange provides the following descriptions:

Description 1: The growth in the capital-exporting countries will be above the steady state.

Description 2: Developed countries will tend to run trade deficits as they export capital.

Description 3: The rate of return on investments will be lower in countries with high capital-to-labor ratios.

Question 12. Which of the following descriptions provided by Team Orange is consistent with Smith's conclusion?

OPTIONS

A	Description 1		Incorrect: growth in the capital-exporting countries will be below the steady state (capital flows out to the capital-poor country).
B	Description 2	✗ Your pick	Incorrect: capital-poor countries run trade deficits while developed countries run trade surpluses as they export capital.
C	Description 3	✓ Correct	Marginal product of capital is lower where capital per worker is high, so returns are lower in high capital-to-labor countries and higher in low capital-to-labor countries.

Trigger “opening up the economy should increase the rate at which countries' capital-to-labor ratios converge” “Description 3: The rate of return on investments will be lower in countries with high capital-to-labor ratios.”

Rule next time: Opening up: capital flows to the low capital-to-labor country (high MPK). The exporter grows BELOW its steady state and runs trade SURPLUSES; the importer grows above steady state and runs deficits. Returns are lower where capital per worker is high.

Key line: High K/L → low MPK → capital leaves.

Mock B · Paper 2 · Q15 You **B** · Answer **C**

PM

VIGNETTE (RELEVANT PART, UNEDITED)

Li uses a two-factor model to analyze the returns of one of the equity portfolios in his fund. He examines how fund returns, RP, are related to surprises in inflation, FINFL, and GDP growth, FGDP, as shown below:

Exhibit 3

Variable	Actual Value (%)	Predicted Value (%)	Factor Sensitivity
Inflation	1.2	0.8	3.0
GDP growth	–3.5	–4.0	0.8

Question 15. Li's two-factor model is best categorized as a:

OPTIONS

A	statistical factor model.		Incorrect: statistical models extract factors (portfolios) from historical returns.
B	fundamental factor model.	✗ Your pick	Incorrect: fundamental factors are company attributes (book-to-price, size, P/E, leverage).
C	macroeconomic factor model.	✓ Correct	The factors are surprises in macroeconomic variables (inflation, GDP growth), so it is a macroeconomic factor model.

Trigger “surprises in inflation, FINFL, and GDP growth, FGDP”

Rule next time: Factors that are SURPRISES in macro variables = macroeconomic factor model; company attributes = fundamental; portfolios extracted from returns = statistical.

Key line: Macro: factor = surprise; fundamental: factor = attribute.

Mock B · Paper 2 · Q16 You **B** · Answer **C**

PM

VIGNETTE (RELEVANT PART, UNEDITED)

Li estimates the expected return of the portfolio to be 1.8% and uses an error term of 0.2%. The actual and expected values of inflation and GDP growth, as well as the estimated portfolio's sensitivities to these factors, are presented in Exhibit 3.

Exhibit 3

Variable	Actual Value (%)	Predicted Value (%)	Factor Sensitivity	
Inflation		1.2	0.8	3.0
GDP growth		–3.5	–4.0	0.8

Question 16. Based on the two-factor model, the return on Li's portfolio is:

OPTIONS

A	1.8%.		Incorrect: this omits the expected return aP (3.0 × 0.4% + 0.8 × 0.5% + 0.2% = 1.8%).
B	2.8%.	✗ Your pick	Incorrect: this uses the actual values instead of the surprises.
C	3.6%.	✓ Correct	RP = 1.8% + 3.0 × (1.2% − 0.8%) + 0.8 × (−3.5% − (−4.0%)) + 0.2% = 1.8% + 1.2% + 0.4% + 0.2% = 3.6%.

Trigger “expected return of the portfolio to be 1.8%”
“Actual Value (%) | Predicted Value (%) | Factor Sensitivity”
Rule next time: R = a + Σ b × (actual − expected) + ε: use SURPRISES, and keep the intercept (expected return) and the error term.

Key line: 1.8 + 3.0 × (1.2 − 0.8) + 0.8 × (−3.5 + 4.0) + 0.2 = 3.6%.

Mock B · Paper 2 · Q18 You **C** · Answer **A**

FI

VIGNETTE (RELEVANT PART, UNEDITED)

Using the spot curve, Li computes forward rates. He then forecasts that in three years the 2-year spot rate will equal 2.20%, the current 2-year spot rate, but that the remainder of the spot rate curve will be consistent with current forward rates. Li evaluates what his forecast implies about the valuation of a 5-year zero-coupon bond today.

Question 18. Given his forecast, Li should consider a 5-year zero-coupon bond today to be:

OPTIONS

A	undervalued.	✓ **Correct**	The forecasted future 2-year spot rate (2.20%) is below the implied forward rate for those last two years, so the bond's cash flows are currently discounted at too high a rate: the bond is undervalued.
B	fairly valued.		Incorrect: a bond is fairly valued only if future spot rates equal today's implied forward rates.
C	overvalued.	✗ **Your pick**	Incorrect: it would be overvalued only if future spot rates exceed the implied forward rates.

Trigger “in three years the 2-year spot rate will equal 2.20%”
“consistent with current forward rates”

Rule next time: If the forecast future spot rate is LOWER than the implied forward rate, today's cash flows are discounted at too high a rate → the bond is undervalued (buy).

Key line: Forecast < forward → undervalued.

Mock B · Paper 2 · Q20 You **A** · Answer **B**

FI

VIGNETTE (RELEVANT PART, UNEDITED)

Li studies the characteristics and effects of yield curve volatility. He uses the 3.25% monthly standard deviation of the 15-year government-bond yield to estimate an annual yield volatility of 11.26%. Li also calculates key rate durations for various maturities. He concludes that the higher key rate durations associated with longer-term rates leads to higher long-term annualized term structure volatility.

Question 20. Are Li's calculation and conclusion related to key rate duration and term structure volatility correct?

OPTIONS

A	Yes	✗ Your pick	Incorrect: the annualized volatility estimate is right but the key rate duration statement is wrong.
B	No, only his annualized volatility estimate is correct	✓ Correct	Annual volatility = 3.25% × √12 = 11.26% is correct. Key rate duration measures price sensitivity to a yield change at one maturity, not how volatile rates are; the volatility term structure typically shows short-term rates more volatile than long-term rates.
C	No, only his conclusion related to key rate duration and term structure volatility is correct		Incorrect: the key rate duration conclusion is the incorrect part.

Trigger "3.25% monthly standard deviation"
"higher key rate durations associated with longer-term rates leads to higher long-term annualized term structure volatility"

Rule next time: Annualise volatility with × √12 (3.25% × 3.464 = 11.26%: correct). Key rate duration is the price sensitivity to ONE maturity's yield; it says nothing about how volatile rates are (short rates are usually more volatile than long rates).

Key line: σ annual = σ monthly × √12.

Mock B · Paper 2 · Q21 You **C** · Answer **B**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Option 1: Borrow directly against the receivables of EUR 50 million;

Option 2: Create a special purpose entity (SPE), invest EUR 10 million in the SPE, have the SPE borrow EUR 40 million, and then use the funds to purchase EUR 50 million of receivables from Acme.

McAndrew considers the potential impact of these options on Acme's consolidated debt-to-assets ratio, assuming that Acme will be able to influence the financial and operating policy of the SPE, and will be expected to absorb any losses that the SPE may incur.

Question 21. Acme's consolidated debt-to-assets ratio would most likely be:

OPTIONS

A	lower under Option 1 compared to Option 2.		Incorrect: Acme controls the SPE (influence over policies and exposure to variable returns), so it consolidates it and the balance sheet looks the same as borrowing directly.
B	the same under either Option 1 or Option 2.	✓ **Correct**	Under IFRS 10 / SIC-12, control (influence over financial and operating policy and exposure to variable returns) requires consolidation, so the consolidated ratio is the same under both options.
C	higher under Option 1 compared to Option 2.	✗ **Your pick**	Incorrect: consolidation makes the two options identical on the balance sheet.

Trigger “will be able to influence the financial and operating policy of the SPE”
“expected to absorb any losses”

Rule next time: Power over the SPE's policies + exposure to its losses = control (IFRS 10) → consolidate: the debt is on the consolidated balance sheet, the same as borrowing directly.

Key line: Control → consolidated ratio identical.

Mock B · Paper 2 · Q25 You **C** · Answer **A**

Equity

VIGNETTE (RELEVANT PART, UNEDITED)

Firm A is a rapidly growing automotive parts distributor that has asked Bala to provide an estimated fair value of its common stock. Bala selects the appropriate valuation model based on the following information about Firm A:

The firm's dividend payout ratio has been stable over the last 10 years and is expected to remain stable, regardless of future profitability.

Planned expansion into new markets will require significant annual capital expenditures, which will result in dividends exceeding free cash flow to equity.

Compared with peers, the firm has an aggressive policy with respect to reported earnings, as well as a lower quality of accounting disclosure.

Question 25. Which of the following valuation models is most appropriate for Firm A?

OPTIONS

A	Free cash flow model	✓ **Correct**	Dividends will exceed FCFE because of the expansion capex, which is when a free cash flow model is most suitable.
B	Residual income model		Incorrect: residual income relies on good-quality accounting, but Firm A's disclosure quality is lower and earnings aggressive.
C	Dividend discount model	✗ **Your pick**	Incorrect: the DDM suits stable dividend policies tied to profitability; the stable payout ratio history is not relevant going forward.

Trigger “dividends exceeding free cash flow to equity”
“lower quality of accounting disclosure”
“regardless of future profitability”

Rule next time: Pick the model from the facts: dividends not tied to profitability and above FCFE → free cash flow model; poor accounting quality rules out residual income; a DDM needs a payout policy tied to profits.

Key line: Dividends > FCFE → FCF model.

Mock B · Paper 2 · Q26 You **C** · Answer **B**

Equity

VIGNETTE (RELEVANT PART, UNEDITED)

Firm B is a regional utilities provider. Firm B's current stock price is $19.00 and Bala uses the dividend discount model to value the stock. In Exhibit 1, Bala forecasts Firm B's earnings for the next three years. He assumes a required return on equity of 8% and a 60% retention ratio. By the end of Year 3, Bala expects Firm B's stock price to be $21.50. ValEwe's approach is to consider a stock to be fairly valued if its price is within 2% of its estimated fair value.

Exhibit 1

	Year 1	Year 2	Year 3
Forecasted EPS	$1.27	$1.62	$1.91

Question 26. Based on the dividend discount model, Bala should consider Firm B's common stock to be:

OPTIONS

A	undervalued.		Incorrect: the $19.00 price is only 1.6% above the $18.70 fair value, within the 2% band.
B	fairly valued.	✓ Correct	Payout = 40%: D1 = 0.51, D2 = 0.65, D3 = 0.76. V0 = 0.51/1.08 + 0.65/1.08² + 0.76/1.08³ + 21.50/1.08³ = $18.70. The price of $19.00 is 1.6% above, within 2%, so fairly valued.
C	overvalued.	✗ Your pick	Incorrect: the price is within the 2% band, so the stock is fairly valued.

Trigger “within 2% of its estimated fair value”
“60% retention ratio”

Rule next time: Payout = 1 − retention = 40% (not 60%): D = 40% × EPS; V0 = PV(dividends) + PV(year-3 price); within ±2% of fair value = fairly valued.

Key line: V0 = $18.70; 19.00 ÷ 18.70 − 1 = +1.6% → fairly valued.

Mock B · Paper 2 · Q32 You **A** · Answer **B**

FI

VIGNETTE (RELEVANT PART, UNEDITED)

The fund owns $5 million of a 7-year senior unsecured bond issued by Gonk Industries. Kehoe recommends the fund buy $5 million of credit protection on Gonk using a 5-year CDS. The CDS has a 5-year senior unsecured bond as its reference obligation. Summary data for the CDS are presented in Exhibit 2.

Two months later, a major accounting scandal is uncovered at Gonk and the firm quickly files for bankruptcy. Gonk has three series of bonds outstanding:

3-year subordinated unsecured bonds trading at 20% of par value

5-year senior unsecured bonds trading at 40% of par value

7-year senior unsecured bonds trading at 30% of par value

Question 32. The cash settlement amount for the fund from the Gonk Industries CDS is closest to:

OPTIONS

A	$3,000,000.	✗ Your pick	Incorrect: the reference (5-year) bond is not the cheapest-to-deliver; the 7-year senior bond is.
B	$3,500,000.	✓ Correct	The payout uses the cheapest-to-deliver deliverable of equal or higher seniority than the reference obligation: the 7-year senior unsecured bond at 30%. LGD = 70%; 70% × $5 million = $3.5 million.
C	$4,000,000.		Incorrect: the 3-year bond (20% of par) is cheaper but subordinated, so it is not deliverable.

Trigger “senior unsecured bond as its reference obligation”
“3-year subordinated unsecured bonds trading at 20% of par value”
“7-year senior unsecured bonds trading at 30% of par value”

Rule next time: CDS payout = LGD of the cheapest-to-deliver bond of the SAME or higher seniority than the reference obligation (any maturity); a cheaper subordinated bond is not deliverable.

Key line: Cheapest senior = 30% → LGD 70% × $5m = $3.5m.

Mock B · Paper 2 · Q38 You **A** · Answer **B**

Alt

VIGNETTE (RELEVANT PART, UNEDITED)

Friso compares two publicly traded real estate companies operating in the retail sector, Perpetua Properties, a REIT, and Rainfall Real Estate, a real estate operating company (REOC). Both companies manage portfolios of retail shopping centers and plan to significantly grow their property portfolios by acquiring undervalued assets.

Question 38. Compared to Rainfall Real Estate, Perpetua Properties most likely has:

OPTIONS

A	a higher effective tax rate.	✗ Your pick	Incorrect: REITs are largely exempt from corporate income tax if they distribute income.
B	a higher dividend payout ratio.	✓ Correct	A REIT must distribute 90-100% of taxable income, so it has a higher payout ratio than a REOC, which can retain earnings.
C	greater flexibility to retain earnings.		Incorrect: a REOC, not a REIT, can retain earnings for acquisitions.

Trigger “Perpetua Properties, a REIT”
“Rainfall Real Estate, a real estate operating company (REOC)”
Rule next time: REIT: exempt from corporate tax if it distributes about 90–100% of taxable income → high payout, little retention; REOC: taxable, can retain earnings for growth.

Key line: REIT payout high; REOC flexible.

Mock B · Paper 2 · Q41 You **A** · Answer **C**

Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

During their initial meeting to set up Kagan's account, Dines proposes using Prism's preferred broker to execute trades. Kagan informs Dines that he wishes to trade through another broker that is owned by a friend. Although Dines knows that Prism's preferred broker would provide better execution, without further discussion she agrees to follow Kagan's wishes and use his broker. Additionally, Dines explains that she will not vote any proxies that would not benefit Kagan's account based on a cost-benefit analysis.

Question 41. Does Dines violate the Standards in the initial meeting with Kagan?

OPTIONS

A	No	✗ Your pick	Incorrect: Dines violates Standard III(A) Loyalty, Prudence, and Care by not telling Kagan that his directed broker may not give best execution.
B	Yes, regarding her proxy voting policy		Incorrect: a cost-benefit approach to proxy voting that is disclosed to the client is acceptable.
C	Yes, regarding the broker for Kagan's account	✓ Correct	In a directed brokerage situation the manager must tell the client he may not be getting best execution; Dines did not, violating Standard III(A).

Trigger “Dines knows that Prism's preferred broker would provide better execution”
“without further discussion she agrees to follow Kagan's wishes”

Rule next time: III(A) directed brokerage: following the client's chosen broker is allowed, but you must TELL him it may cost best execution. A disclosed cost-benefit proxy policy is fine.

Key line: Directed broker → disclose possible loss of best execution.

Mock B · Paper 2 · Q42 You **C** · Answer **B**

Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

At a subsequent meeting, Dines informs Kagan of the premium services offered by her firm at an additional cost, which include direct access to senior research professionals. Kagan decides not to purchase the premium services. A few days later Prism publishes a report changing its recommendation on Cromwell Inc., a cybersecurity company, to a sell. Shortly after publication Dines sets up calls for her premium clients to have detailed discussions with the senior analyst who authored the report, but she does not set up such a call for Kagan. The next day, Kagan calls Dines and asks her to buy Cromwell shares for his account. Dines knows that Kagan is an expert in the cybersecurity sector from their previous interactions. She proceeds to place the order, without informing him of the firm's recent recommendation change.

Question 42. Does Dines violate the Standards in her actions related to Cromwell?

OPTIONS

A	No		Incorrect: Dines violated Standard III(B) Fair Dealing by not telling Kagan about the changed recommendation.
B	Yes, because of her actions related to the buy order for Kagan's account	✓ Correct	If a client places an order contrary to a current recommendation, the client should be told of the changed recommendation before the order is accepted. Dines did not tell Kagan the rating had changed to sell.
C	Yes, because of her actions related to setting up calls with the senior analyst	✗ Your pick	Incorrect: fair dealing means not equally; premium clients pay for the calls and the calls came after publication, so there is no violation.

Trigger “Prism publishes a report changing its recommendation on Cromwell Inc., a cybersecurity company, to a sell”
“without informing him of the firm's recent recommendation change”

Rule next time: III(B) Fair dealing: if a client orders against a current recommendation, tell him about the change before accepting the order. Paid premium calls after publication are fine (fair, not equal).

Key line: Order against recommendation → inform first.

Mock B · Paper 2 · Q43 You **C** · Answer **A**

Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

Several months later, Kagan meets with Dines to discuss how his account has performed relative to other accounts managed with the same strategy. Dines creates a performance presentation that shows the past five years of return data for the top and bottom deciles of all Prism accounts managed with the same strategy. The presentation also includes a disclosure stating that the performance is shown before taxes and gross of fees.

Question 43. Dines violates the Standards in her performance presentation for Kagan with regard to:

OPTIONS

A	the account sample only.	✓ Correct	Showing only the top and bottom deciles is not fair, accurate and complete (Standard III(D)); a weighted composite of similar portfolios should be used.
B	the disclosure of taxes and fees only.		Incorrect: the disclosure that returns are gross of fees and before taxes is adequate.
C	both the account sample and the disclosure of taxes and fees.	✗ Your pick	Incorrect: only the account sample is a violation; the disclosure is fine.

Trigger “the past five years of return data for the top and bottom deciles of all Prism accounts” “shown before taxes and gross of fees”

Rule next time: III(D): present a weighted composite of similar portfolios, not selected deciles or accounts; disclosing 'gross of fees, before taxes' is adequate.

Key line: Composite = fair, accurate, complete.